

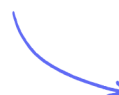
Multiple Choice Questions

Alkanes

Alkanes - Introduction / Alkanes as Fuels / Combustion of Alkanes / Alternative Fuels / Free Radicals & Fission / The Free Radical Substitution Mechanism

Easy (3 questions)	/3
Medium (5 questions)	/10
Total Marks	/13

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Easy Questions

1 All alkanes have the same

- A. empirical formula
- B. general formula
- C. molecular formula
- D. structural formula

(1 mark)

2 A single molecule of decane, $C_{10}H_{22}$, is cracked.

Which of these mixtures could **not** be formed?

- A. pentene and pentane
- B. ethene, butene and butane
- C. propene, propane and butene
- D. hexene and propane

(1 mark)

3 Which pollutant **cannot** form when alkane fuels are burned in car engines?

- A. hydrogen chloride
- B. sulfur dioxide
- C. carbon particulates
- D. carbon monoxide

(1 mark)

Medium Questions

1 (a) Ethane reacts with bromine in the presence of ultraviolet radiation.

What is the equation for the reaction?

- A.** $\text{C}_2\text{H}_6 + \text{Br}_2 \rightarrow \text{C}_2\text{H}_4\text{Br}_2 + \text{H}_2$
- B.** $\text{C}_2\text{H}_6 + \text{Br}_2 \rightarrow \text{C}_2\text{H}_5\text{Br} + \text{HBr}$
- C.** $\text{C}_2\text{H}_6 + \text{Br}_2 \rightarrow 2\text{CH}_3\text{Br}$
- D.** $\text{C}_2\text{H}_6 + \text{Br}_2 \rightarrow \text{CH}_4 + \text{CH}_2\text{Br}_2$

(1 mark)

(b) The ultraviolet radiation is needed for

- A.** homolytic breaking of a Br-Br bond
- B.** heterolytic breaking of a Br-Br bond
- C.** homolytic breaking of a C-H bond
- D.** heterolytic breaking of a C-H bond

(1 mark)

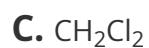
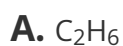
2 (a) Methane reacts with excess chlorine in UV light.

Which process occurs in the initiation step?



(1 mark)

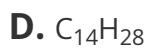
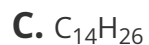
(b) Which of these molecules could **not** be formed in a termination step?



(1 mark)

3 When $\text{C}_{20}\text{H}_{42}$ is cracked, each molecule produces one molecule of ethene, one molecule of butane and two molecules of hydrocarbon **E**.

What is the molecular formula of **E**?



(1 mark)

4 Petrol, bioethanol and hydrogen are fuels.

All three of these fuels

- A.** burn to produce greenhouse gases
- B.** are overall carbon neutral
- C.** are overall sustainable
- D.** biodegrade rapidly

(1 mark)

5 (a) Cyclopentane undergoes free radical substitution with bromine.

a)

Which of these is an overall equation for this reaction?

- A. $\text{C}_5\text{H}_8 + \text{Br}_2 \rightarrow \text{C}_5\text{H}_8\text{Br}_2$
- B. $\text{C}_5\text{H}_{10} + \text{Br}_2 \rightarrow \text{C}_5\text{H}_{10}\text{Br}_2$
- C. $\text{C}_5\text{H}_{10} + \text{Br}_2 \rightarrow \text{C}_5\text{H}_8\text{Br}_2 + \text{H}_2$
- D. $\text{C}_5\text{H}_{10} + \text{Br}_2 \rightarrow \text{C}_5\text{H}_9\text{Br} + \text{HBr}$

(1 mark)

(b) Which statement is **not** correct about this reaction system?

- A. only the initiation step involves homolytic bond fission
- B. only some bromine is converted to free radicals in the initiation step
- C. propagation forms more product than termination
- D. further substitution reactions are likely to occur

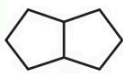
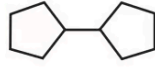
(1 mark)

(c) Which free radical is **least** likely to form in a propagation step in this reaction system?

- A. $\text{C}_5\text{H}_9^\bullet$
- B. $\text{Br}^\bullet$
- C. $\text{C}_5\text{H}_8\text{Br}^\bullet$
- D. $\text{H}^\bullet$

(1 mark)

(d) Which alkane could be formed in a termination step in this reaction system?

- ☒ **A** 
- ☐ **B** 
- ☐ **C** 
- ☐ **D** 

(1 mark)